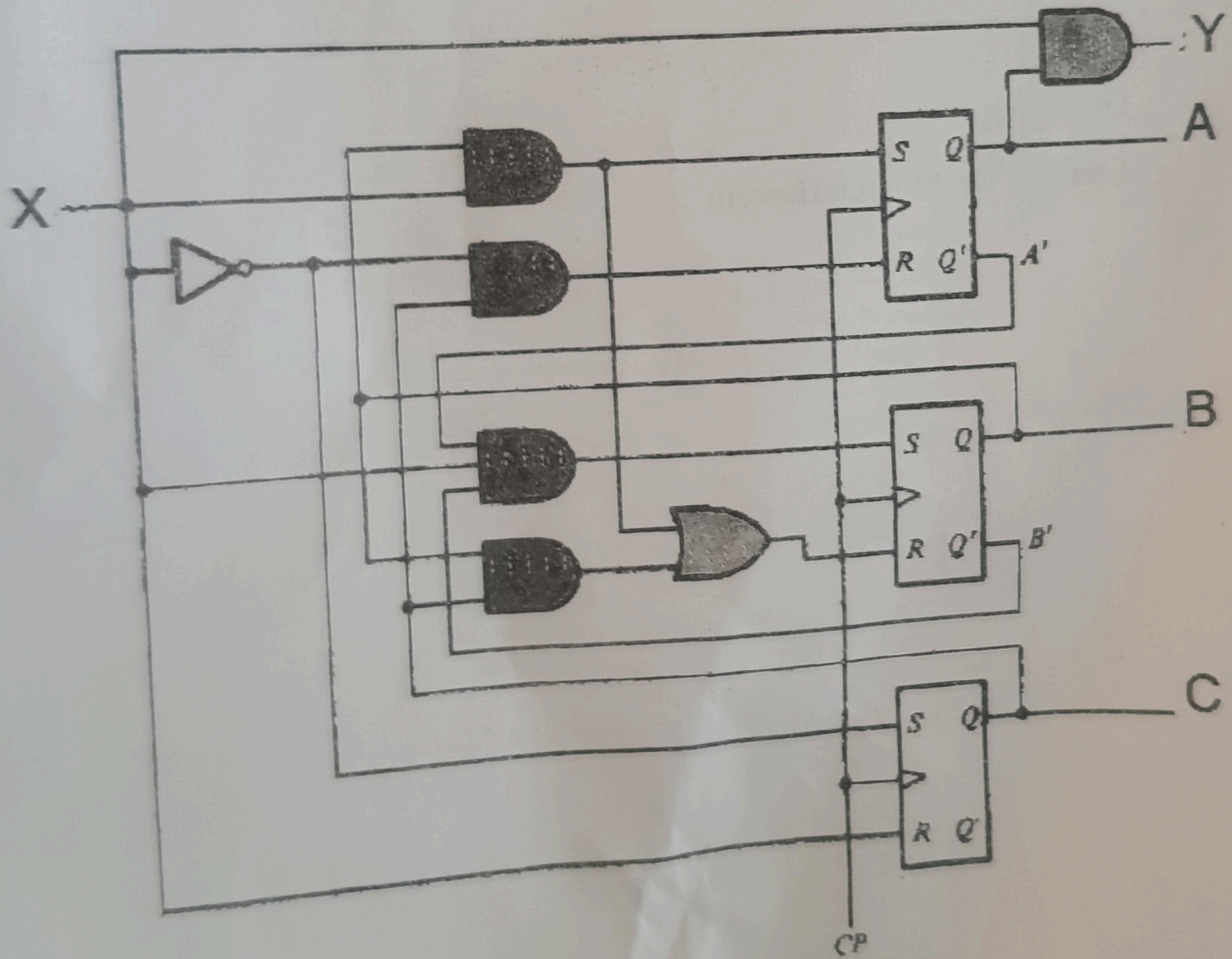


Question 1:

- With drawing explain the differences between combinational and sequential logic?
- Analyze the following clocked sequential circuit. Obtain Y ; A ; B ; C



Question 2

How do you design a logic circuit that has three inputs, A, B, and C, and whose output will be high only when majority of the inputs are high?

Question 3:

What are the differences between Decoder and Encoder?
Explain with drawing

Question 4:

Implement the following function with a multiplexer:

$$F(a, b, c) = \sum (1, 3, 5, 7, 8, 10, 12, 15)$$

Question 5:

Design T flip-flop with using JK flip flop?

Good luck

Lecturer: Dr. Mohsen Hussein